

International Corporate Finance

2023.09.14.

Exchange rate behaviour and the
international long term financing

Dr. Judit Burucs

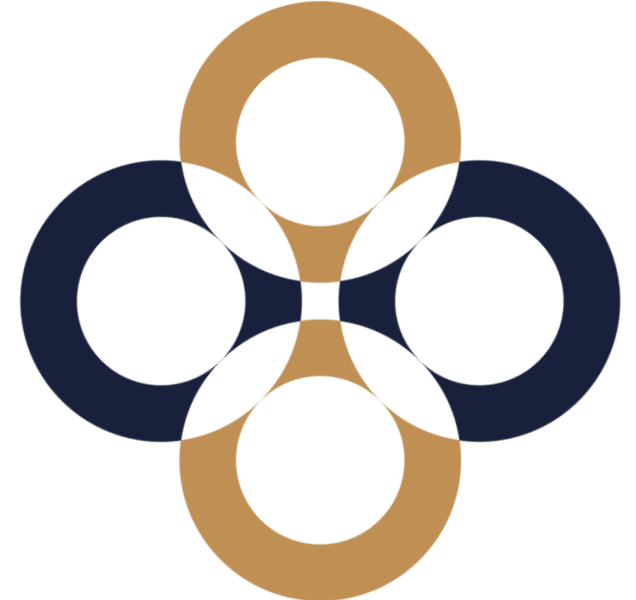
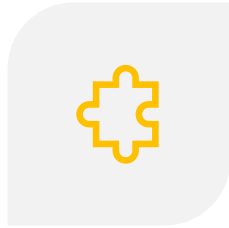


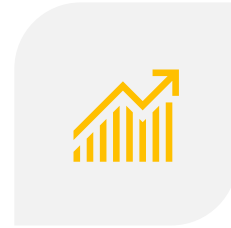
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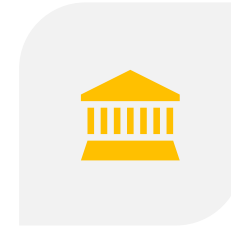
INTERNATIONAL
FINANCIAL
ENVIRONMENT



Exchange Rate
Behaviour



International Corporate
Finance



-Long term Financing
Cost of Capital

Lecturer- Class

Dr. Judit Burucs gained from her experience:

- Senior lecturer in Corvinus University of Budapest
- As a director and head of the Strategic, Innovation and Risk Management (Hungarian Post),
- Managing Director and Head of Group Level Country, Counterparty, Liquidity and Market Risk Department, Hungary (OTP Group),
- Program/Project Manager of Russia Banking Advisory Project, (IFC, Moscow).
- She was also member of
 - the Supervisory Board at TakarekBank, and FHB Bank in Hungary
 - the Board of Directors at Magyar Posta Befektetési Zrt. (Investment Company) and at Díjbeszedő Faktorház Zrt. (Factoring company), as well as OTP Voluntary Pension Fund
- Manager at KPMG

Contacts:

Judit.burucs@uni-corvinus.hu

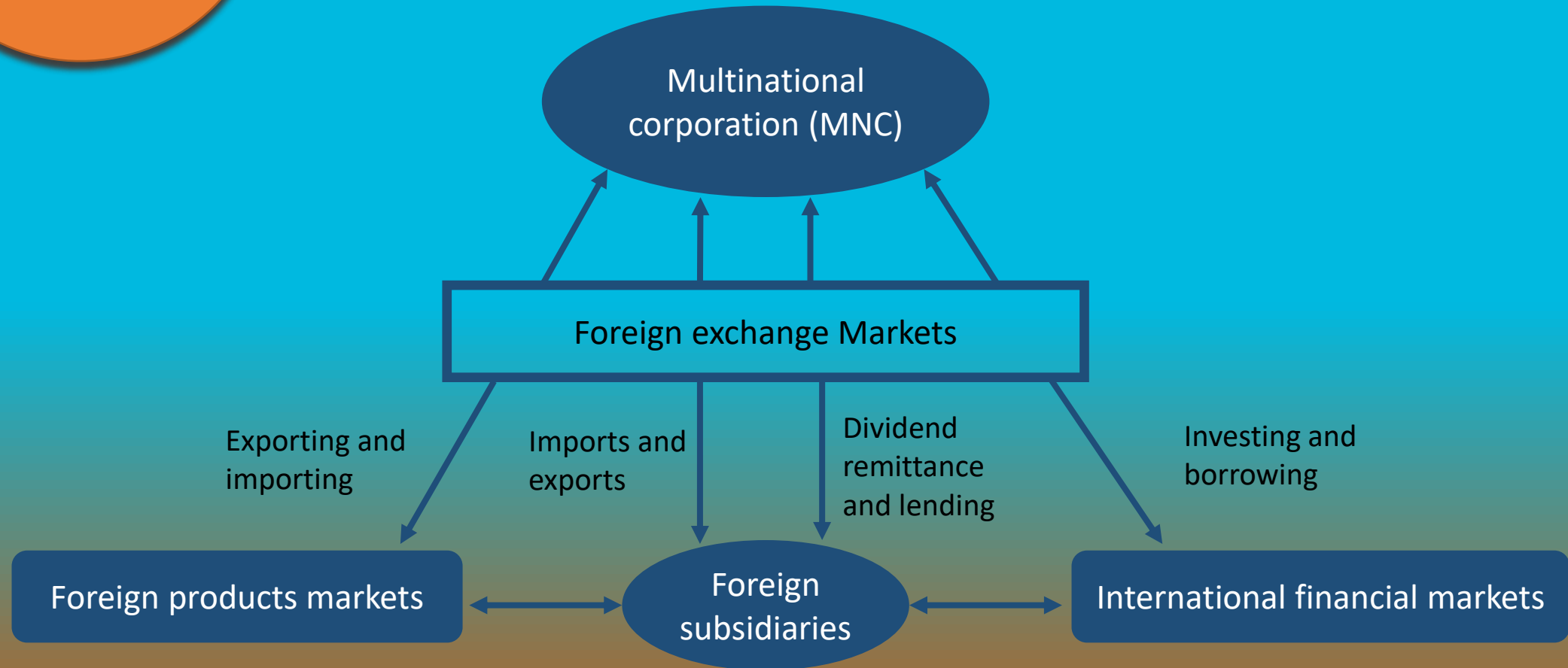
juditburucs@gmail.com

C420

+36707080269

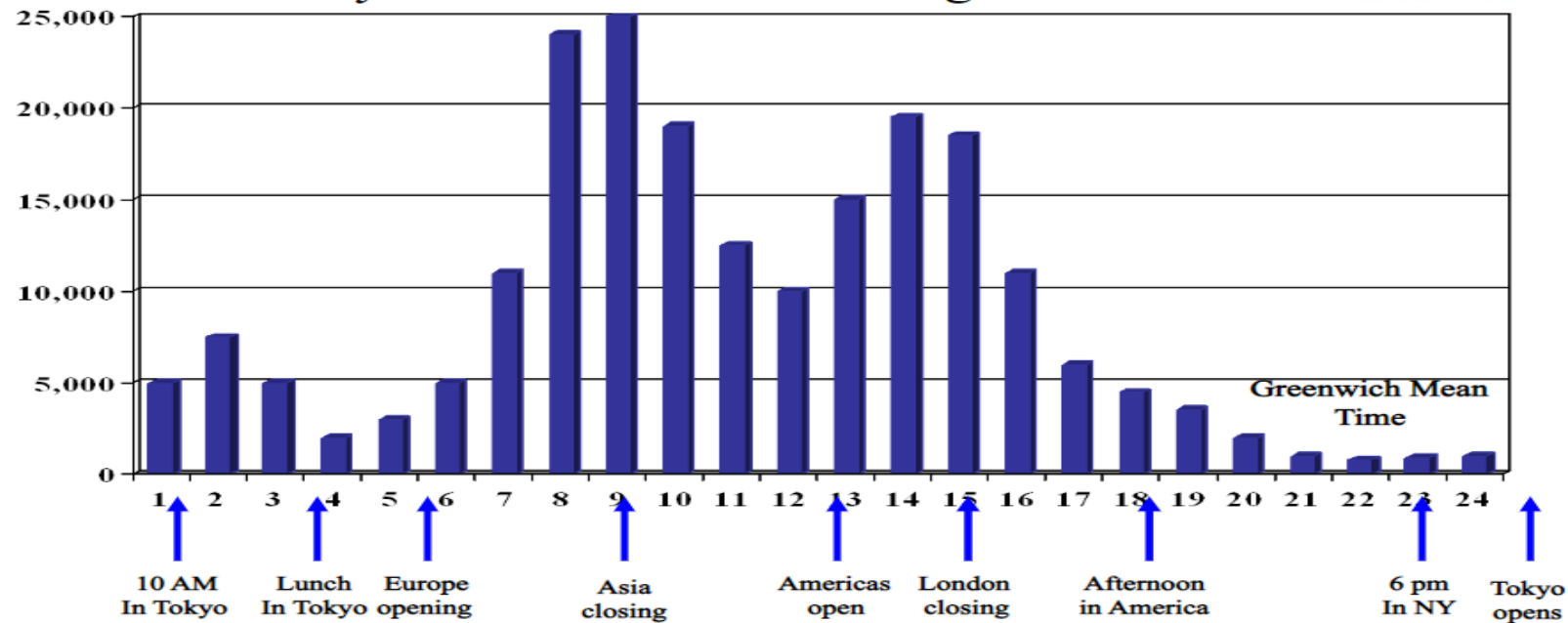
Part 1

The International Financial Environment



Currency market

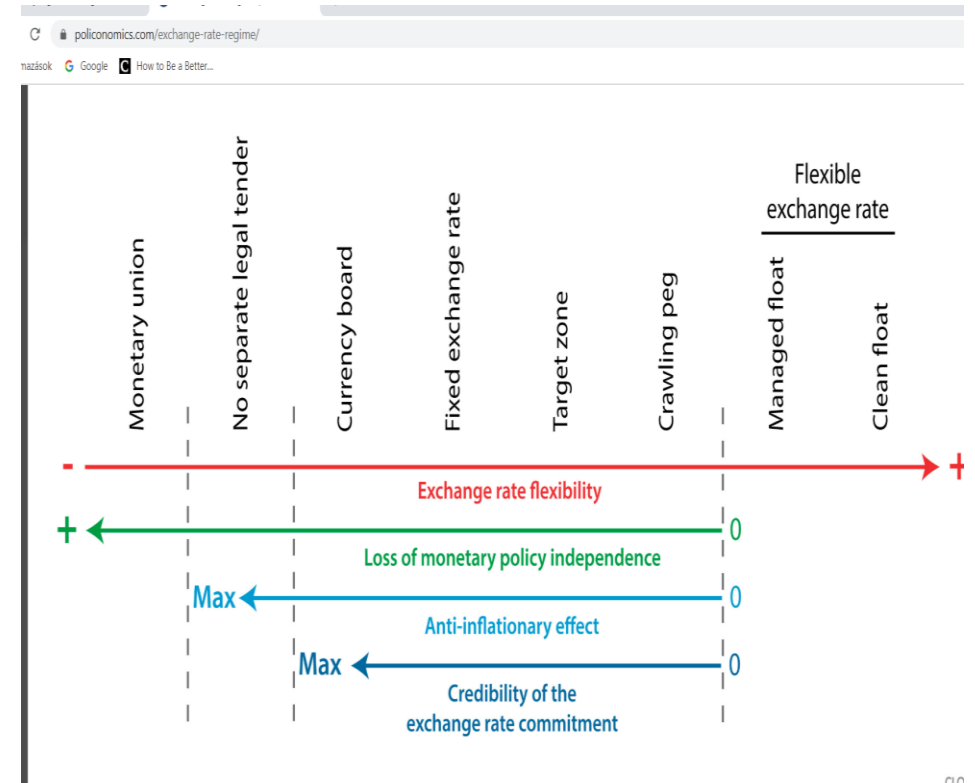
Exhibit 4.1 Measuring Foreign Exchange Market Activity:
Average Electronic Conversations Per Hour.
volume of currency transactions across the globe



Source: Federal Reserve Bank of New York, "The Foreign Exchange Market in the United States," 2001, www.ny.frb.org.

An exchange rate regime

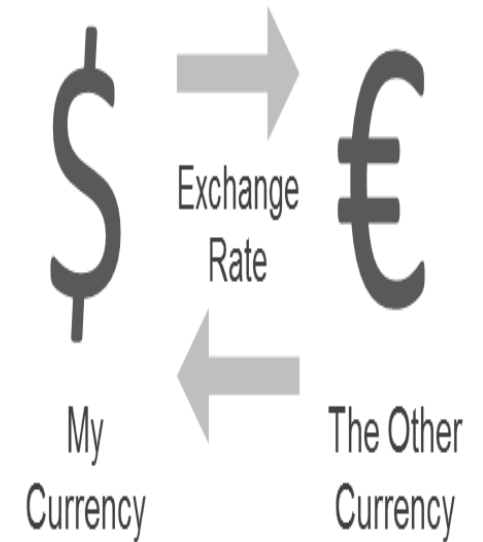
- An exchange rate regime is the system that a country's monetary authority, - generally the central bank-, adopts to establish the exchange rate of its own currency against other currencies. Each country is free to adopt the exchange-rate regime that it considers optimal, and will do so using mostly monetary and sometimes even fiscal policies.
- The distinction amongst these exchange rates regimes is generally just made between fixed and flexible exchange rate regimes, but we find there are many other different regimes, some of which are in between these extreme cases:
- – Monetary Union, with a shared currency, such as the Eurozone;
- – No separate legal tender, where the use of the currency of another country takes place;
- – Currency Board, an explicit agreement on a fixed exchange rate between two or more currencies;
- – Target zone arrangement, where the exchange rate is allowed to fluctuate within certain bands;
- – Crawling Peg, with a periodically adjusted exchange rate;
- – Managed (dirty) float, a flexible exchange rate regime with some government intervention;
- – Free (clean) float, the exchange rate is market determined.



<https://policonomics.com/exchange-rate-regime/>

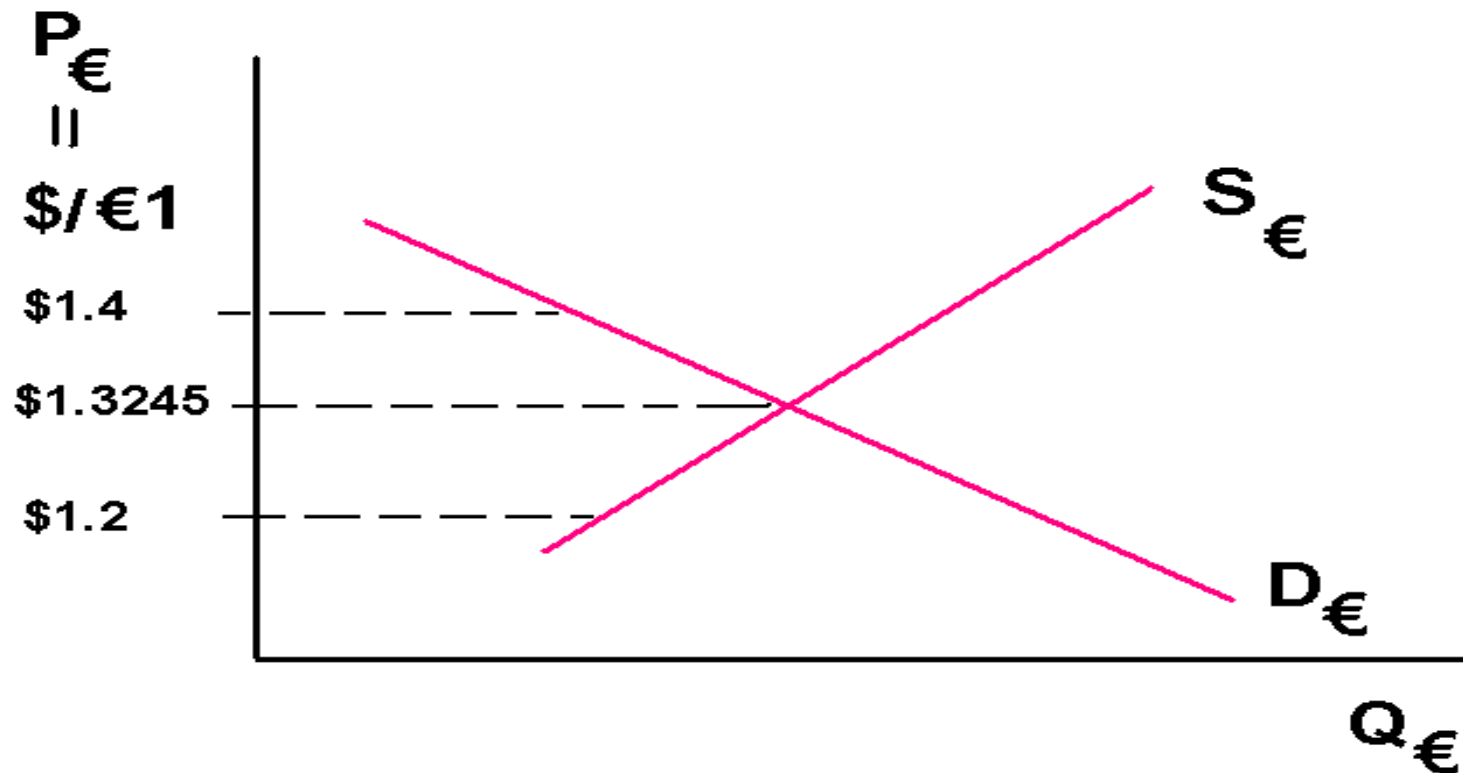
Currency market-Spot market and forward market

- The Currency Market: where money denominated in one currency is bought and sold with money denominated in another currency
- Location:OTC-type: no specific location
- The most common type of foreign exchange transaction is for immediate exchange at the so-called spot rate (**immediate is interpreted as being within two days**).
- The market where these transactions occur is known as the spot market. The average daily foreign exchange trading: *About 27% of transactions are US dollar–euro transactions, 13% US dollar–yen and 12% US dollar–British pound.*
- Forwards are executed between banks or between a bank and a customer; futures are done on an exchange, which is a party to the transaction.
- Swap market: spot and forward together



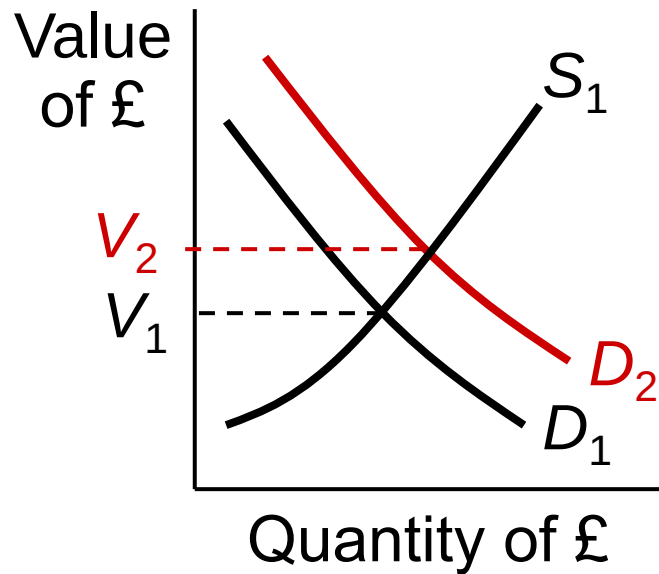
Factor Affecting the Exchange rates

$$\Delta FX = f(\Delta \text{Infl.diff.}, \Delta i \text{ diff.}, \Delta \text{GDP diff.}, \Delta \text{Gov.Contr.}, \Delta \text{Exp(fx)})$$

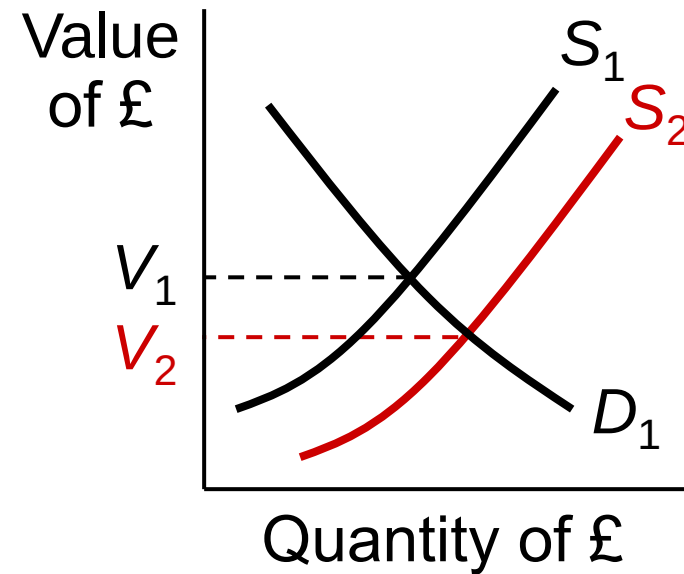


Effects of Bank of England (BoE) Intervention in the Foreign Exchange Market

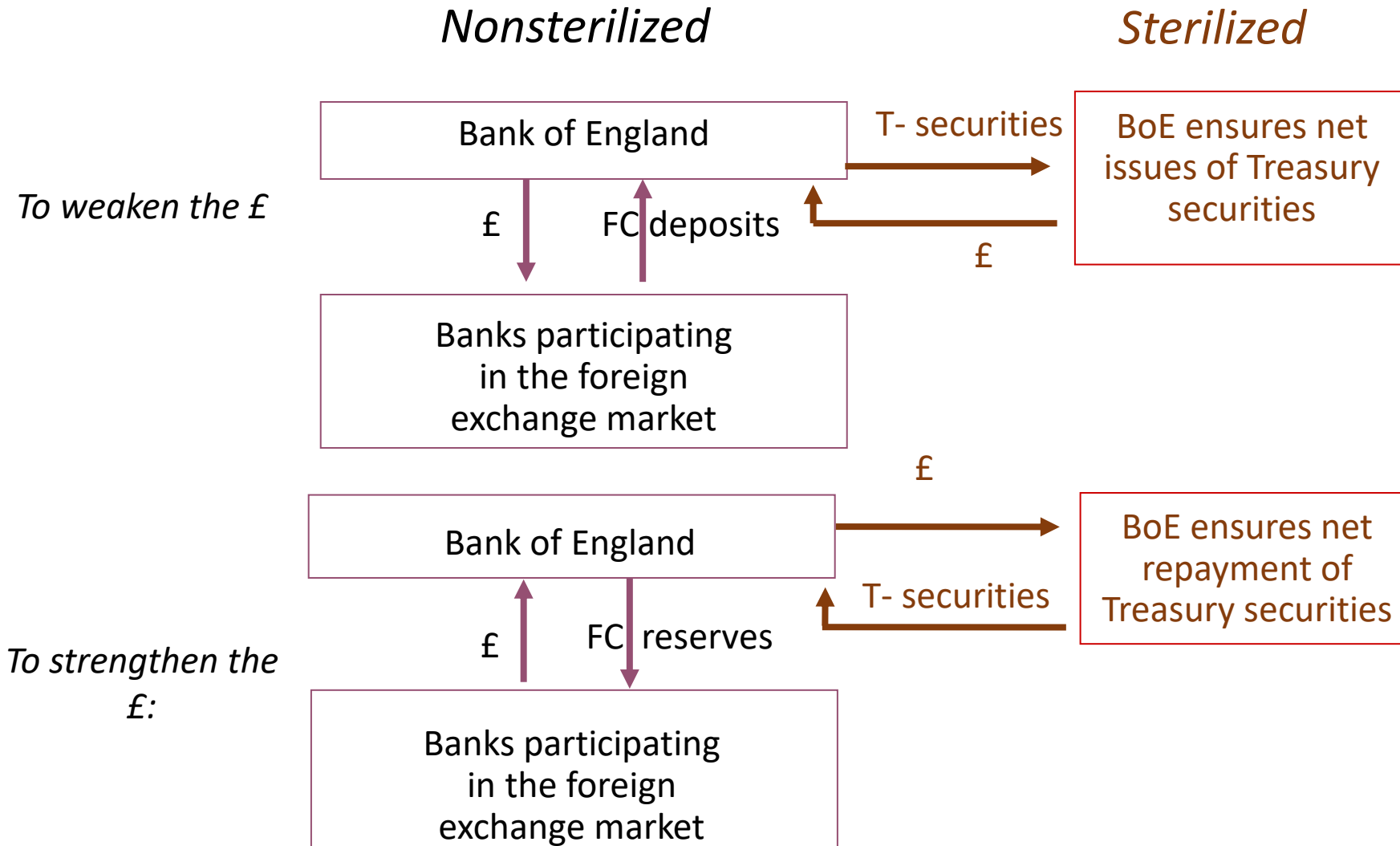
BoE buys £'s with reserves of FC to strengthen the £



BoE sells £'s for FC to weaken the £



Nonsterilized v. Sterilized Intervention



Problems

- 1) Intervention with Euros:** Assume that Belgium, one of the European countries that uses the euro as its currency, would prefer that its currency depreciate against the dollar. Can it apply central bank intervention to achieve this objective? Explain.
- 2) Indirect Intervention.** How can a central bank use indirect intervention to change the value of a currency?

Foreign Exchange Rates and Quotations

Information for travelers

Euro to US Dollar Exchange Rate

1 € = \$ 1,3220 ▲ 0.25%

[Learn to trade currencies with Easy-Forex.com - Open a demo account for free!](#)

Major currency exchange rates in real-time

	EUR	USD	GBP	CHF	CAD	AUD	JPY	SEK	TRY
EUR =	-	1.322	0.8194	1.2075	1.3172	1.2741	113.72	8.6003	2.363
USD =	0.7564	-	0.6197	0.9133	0.9963	0.9637	86.016	6.5046	1.7873
GBP =	1.2204	1.6134	-	1.4737	1.6076	1.555	138.78	10.495	2.8837
CHF =	0.8281	1.0948	0.6785	-	1.0909	1.0552	94.19	7.1216	1.9568
CAD =	0.7591	1.0036	0.622	0.9166	-	0.9672	86.32	6.5284	1.7938
AUD =	0.7848	1.0375	0.643	0.9476	1.0338	-	89.213	6.749	1.8544
JPY =	0.0087	0.0116	0.0072	0.0106	0.0115	0.0112	-	0.0756	0.0207

- For example, the exchange rate between US dollars and the Swiss franc is normally stated:
 - SF 1.6000/\$ (European terms)
- However, this rate can also be stated as:
 - \$0.6250/SF (American terms)
- Excluding two important exceptions, most interbank quotations around the world are stated in European terms.

THE SPOT MARKET quotation

Percent Spread Formula (PS):

$$PS = \frac{Ask - Bid}{Ask} \times 100$$

Market participant:

- Speculator
- Hedger
- Arbitrager

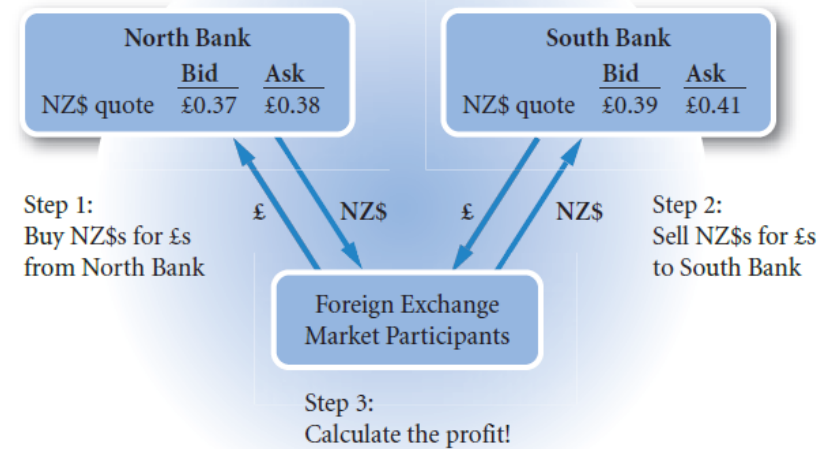
- Interbank quotations are given as a bid and ask (also referred to as offer).
- A *bid* is the price (i.e. exchange rate) in one currency at which a dealer will buy another currency.
- An ask is the price (i.e. exchange rate) at which a dealer will sell the other currency.
- Dealers bid (buy) at one price and ask (sell) at a slightly higher price, making their profit from the spread between the buying and selling prices.
- A bid for one currency is also the offer for the opposite currency.

Locational Arbitrage: Bid and ask

EXHIBIT 7.1 Currency quotes for locational arbitrage example

	Akron Bank			Zyn Bank	
	Bid (bank buys dollars)	Ask (bank sells dollars)		Bid (bank buys dollars)	Ask (bank sells dollars)
Dollar quote	£0.61	£0.62	Dollar quote	£0.62	£0.63

EXHIBIT 7.2 Locational arbitrage

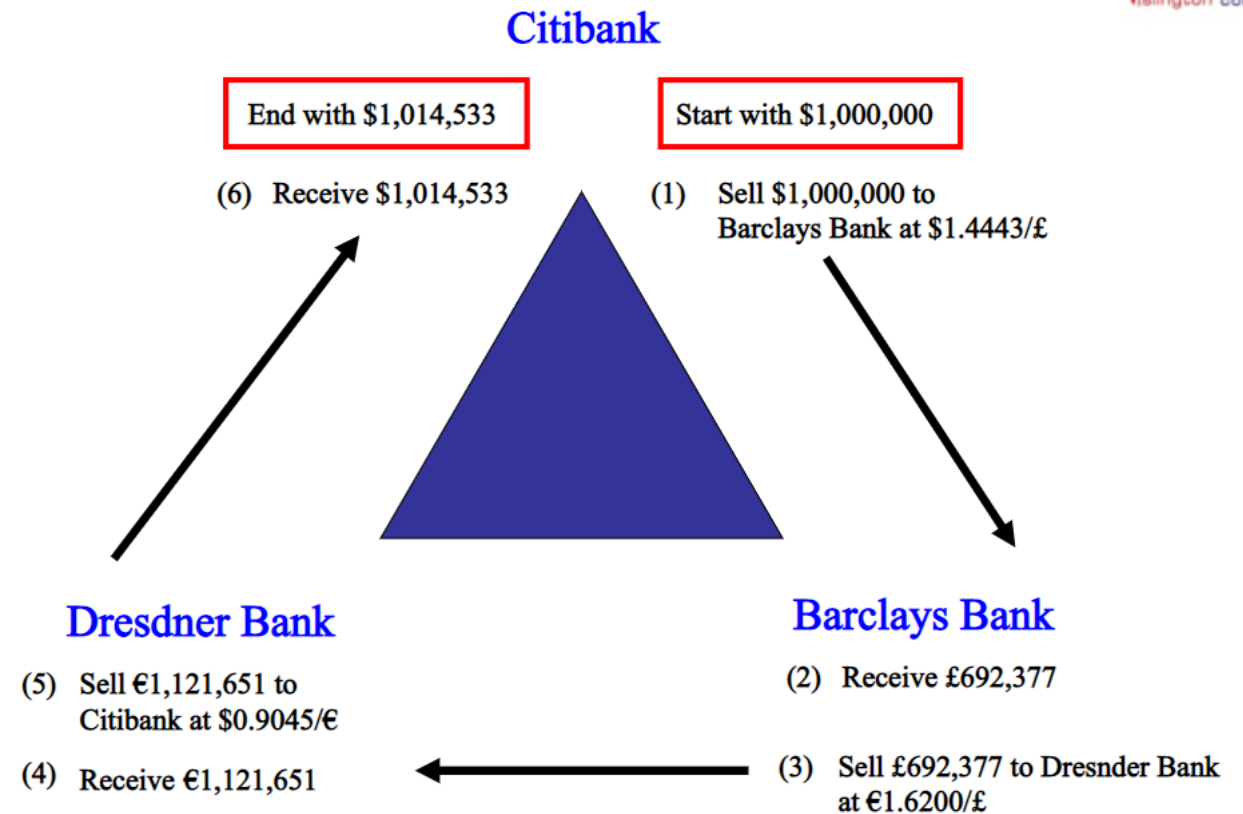


Triangular Arbitrage: Cross currency

<https://slideplayer.com/slide/6359476/>

- Citibank quote - \$/€ \$0.9045/€
- Barclays quote - \$/£ \$1.4443/£
- Dresdner quote - €/£ €1.6200/£
- Cross rate calculation:

$$\frac{\$1.4443/£}{\$0.9045/€} = € 1.5968/£$$



Problem: Triangular Arbitrage.

Assume the following information:

Quoted Price

Value of Canadian dollar in U.S. dollars	\$.90
Value of New Zealand dollar in U.S. dollars	\$.30
Value of Canadian dollar in New Zealand dollars	NZ\$3.02

Given this information, is triangular arbitrage possible?

If so, explain the steps that would reflect triangular arbitrage, and compute the profit from this strategy if you had \$1,000,000 to use.

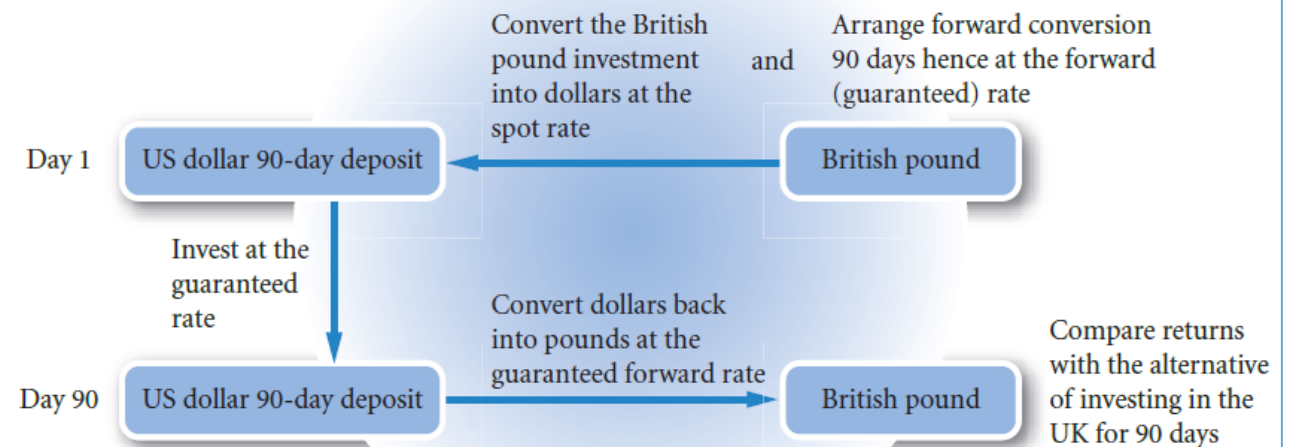
What market forces would occur to eliminate any further possibilities of triangular arbitrage?

Covered interest rate arbitrage: spot and forward

As a UK currency trader, you desire to capitalize on relatively high rates of interest in the United States compared with the United Kingdom and have funds available for 90 days. The interest rate is certain; only the future exchange rate from dollars into pounds is uncertain. You can use a forward sale of dollars to guarantee the rate at which you can exchange dollars for pounds at a future point in time (see Exhibit 7.6). The strategy is as follows:

- 1 On day 1, convert your British pounds into dollars and set up a 90-day deposit account in a US bank. Work out value of the account after 90 days.
- 2 On day 1, engage in a 90-day forward contract to sell the dollars accumulated in step 1 for pounds. The arbitrage deal is now complete and the profit guaranteed.
- 3 In 90 days, when the deposit matures, fulfil the 90-day forward contract and convert the dollars back into pounds at the agreed forward rate.

EXHIBIT 7.6 Covered interest arbitrage



Problem

EXAMPLE

Suppose that a holder of euros is considering investing in the US to exploit the higher interest rate. A US bank has quoted the following details:

	Bid Quote	Ask Quote
Euro spot	\$1.28	\$1.30
Euro one-year forward	\$1.30	\$1.31
	Deposit Rate	
Interest rate on dollars	7.5%	
Interest rate on euros	6.5%	

You have 100 000 euros to invest for one year. Would you benefit from engaging in covered interest arbitrage?

Covered Interest Arbitrage. Assume the following information:

Quoted Price

Spot rate of Canadian dollar \$.80

90-day forward rate of Canadian dollar \$.79

90-day Canadian interest rate 4%

90-day U.S. interest rate 2.5%

Given this information, what would be the yield (percentage return) to a U.S. investor who used covered interest arbitrage?

(Assume the investor invests \$1,000,000.) What market forces would occur to eliminate any further possibilities of covered interest arbitrage?

Forward rate quotation

- Forward rates are typically quoted in terms of points.
- A forward quotation is expressed in points is not a foreign exchange rate as such.
- Rather, it is the *difference* between the forward rate and the spot rate.
- Forward quotations may also be expressed as the percent-per-annum deviation from the spot rate.
- This method of quotation facilitates comparing premiums or discounts in the forward market with interest rate differentials.

- For quotations expressed in foreign currency terms (Indirect quotations) the formula becomes:

$$f^{\%} = \frac{\text{Spot} - \text{Forward}}{\text{Forward}} \times \frac{360}{n} \times 100$$

- For quotations expressed in home currency terms (Direct quotations) the formula becomes:

$$f^{\%} = \frac{\text{Forward} - \text{Spot}}{\text{Spot}} \times \frac{360}{n} \times 100$$

Foreign Exchange Rates and Quotes



- Measuring a change in the spot rate for quotations expressed in home currency terms (direct quotations):

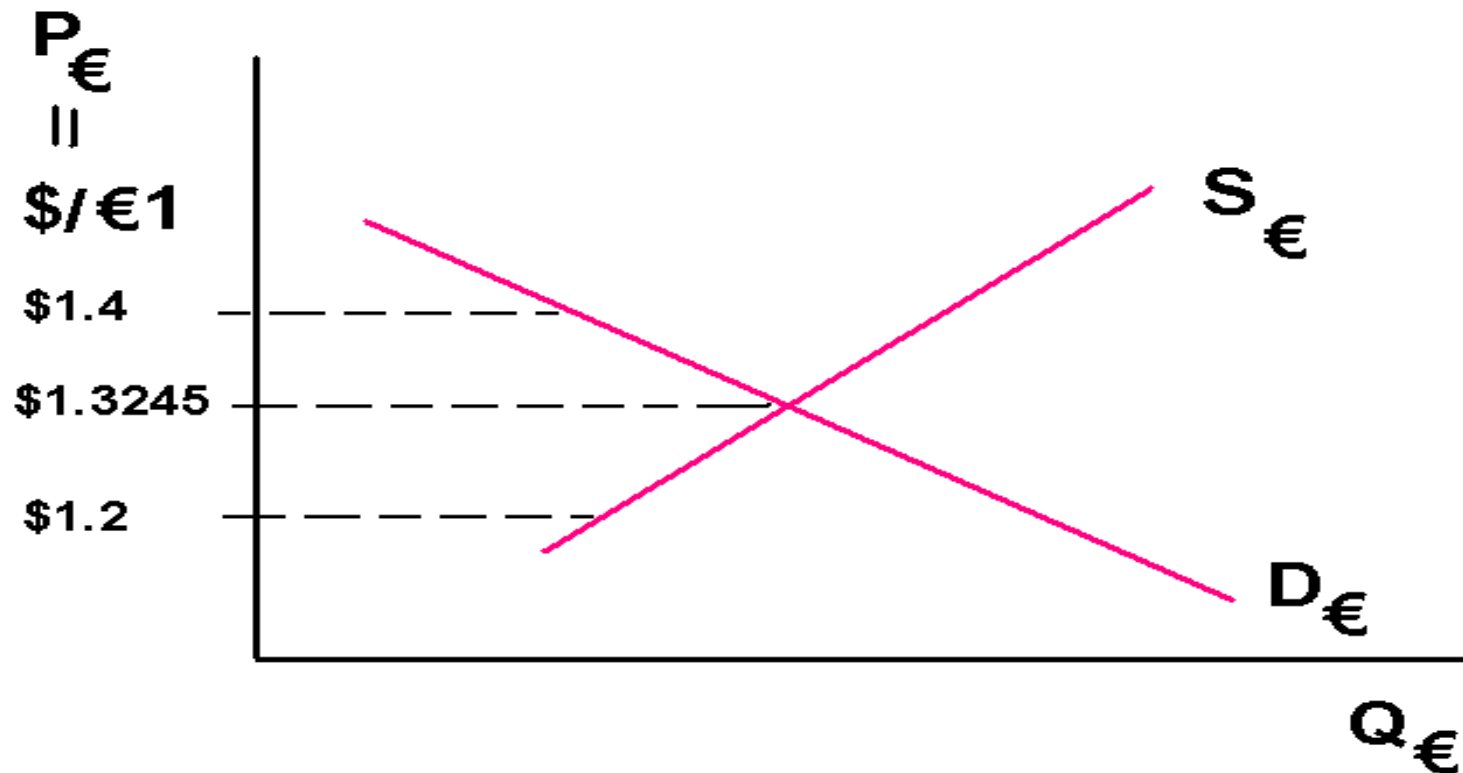
$$\% \Delta = \frac{\text{Ending rate} - \text{Beginning Rate}}{\text{Beginning Rate}} \times 100$$

- Quotations expressed in foreign currency terms (indirect quotations):

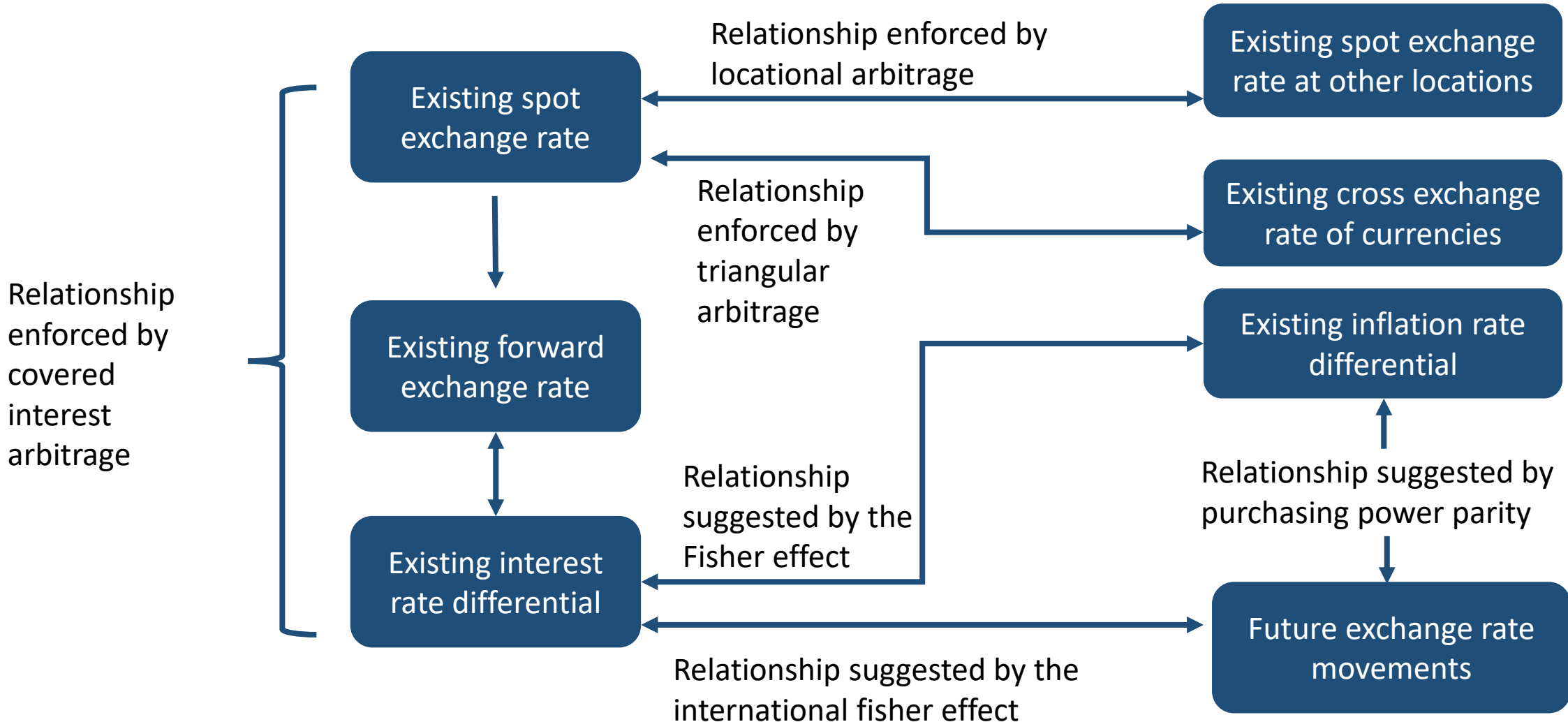
$$\% \Delta = \frac{\text{Beginning Rate} - \text{Ending Rate}}{\text{Ending Rate}} \times 100$$

Factor Affecting the Exchange rates

$$\Delta FX = f(\Delta \text{Infl.diff.}, \Delta i \text{ diff.}, \Delta \text{GDP diff.}, \Delta \text{Gov.Contr.}, \Delta \text{Exp(fx)})$$



Exchange Rate Behaviour





What Is Purchasing Power Parity?

A theoretical exchange rate that allows you to buy the same amount of goods and services in every country

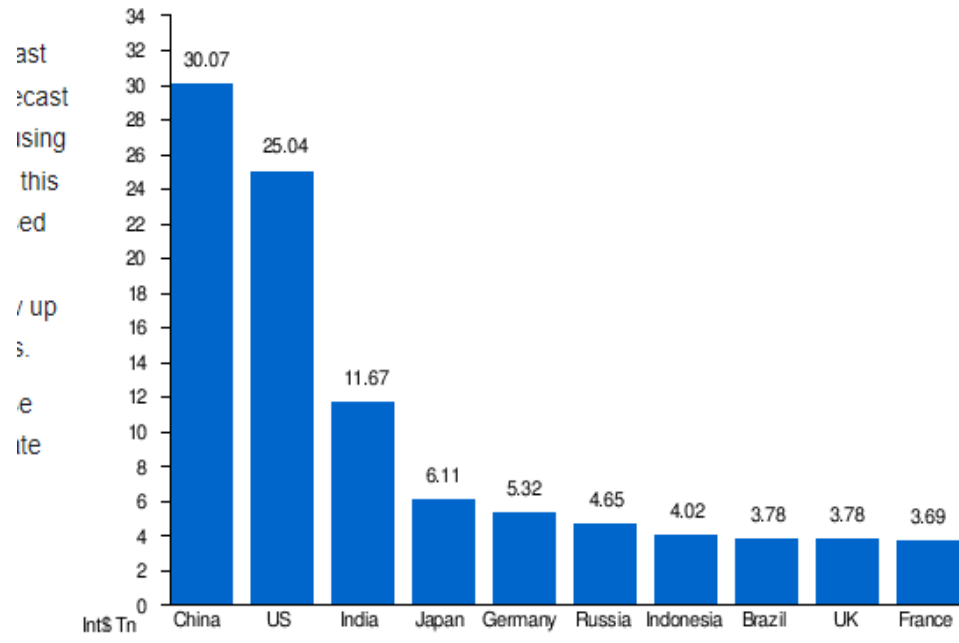


Example: If you want to live cheap, and you can move to any country in the world, compare prices of a Big Mac



Government agencies use it to compare the output of countries that use different exchange rates

PPP



Purchasing
power parity
Formula

$$\text{Purchasing power parity} = \frac{\text{Cost of good X in currency 1}}{\text{Cost of good X in currency 2}}$$

Example

OECD PPP Exchange Rates and Actual Rates

FX	2005	2006	2007	2008	2009	2010
USD/JPY	110.22	116.30	117.75	103.36	93.57	87.78
PPP:	129.55	124.61	120.16	116.84	115.35	111.38
USD/CHF	1.2452	1.2538	1.2004	1.0831	1.0881	1.0429
PPP:	1.7427	1.6593	1.5997	1.5505	1.5313	1.5104
USD/CNY	8.1943	7.9734	7.6075	6.9487	6.8314	6.7703
PPP:	3.448	3.465	3.622	3.821	3.764	3.946
EUR/USD	1.2436	1.2545	1.3687	1.4648	1.3892	1.3244
PPP:	1.1670	1.2035	1.2157	1.2383	1.2505	1.2357
GBP/USD	1.8182	1.8400	2.0001	1.8384	1.5578	1.5452
PPP:	1.5719	1.5949	1.5506	1.5638	1.5570	1.5349
AUD/USD	0.7637	0.7530	0.8368	0.8388	0.7799	0.9173
PPP:	0.7203	0.7095	0.7008	0.6761	0.6887	0.6611

Purchasing Power Parity (PPP)

- When a country's inflation rate rises relatively to that of another country, decreased exports and increased imports depress the high-inflation country's currency. **Purchasing power parity (PPP)** theory attempts to quantify this inflation – exchange rate relationship.
 - The **absolute form of PPP** is an extension of the law of one price. It suggests that the prices of the same products in different countries should be equal when measured in a *common* currency. Generally does not hold prices differences.
 - The **relative form of PPP** accounts for market imperfections like transportation costs, tariffs, and quotas. It states that the differences should be constant.

Rationale behind PPP Theory – based on trade

Suppose prices between countries differ by an average ratio r

(e.g. $r = 0.1$ if home goods 10% more expensive, -0.05 if 5% cheaper, and so on.)

$$\text{\$value goods} \times S \times (1+r) = \text{\pounds value goods}$$

S = price of a
foreign currency
unit

Suppose UK (home) inflation > US inflation.

⇒ ↑ UK imports from US (more demand for \$s) and ↓ UK exports to US (less demand for £s)

⇒ Upward pressure is placed on the value of the \$.

Rationale behind PPP Theory – based on trade

S = price of a
foreign currency
unit

Suppose UK (home) inflation > US inflation...

This *shift* in consumption and the \$'s *appreciation* will continue against the £ until S changes such that...

In the UK buyers find that US goods are once again restored to their previous differences...

1. In the UK: $\text{price}_{\text{US goods}} \times S \times (1+r) \approx \text{price}_{\text{UK goods}}$

In the US buyers find that UK goods \$ imported price have declined so that they are now restored to their previous differences.

2. In the US : $\text{price}_{\text{UK goods}} \times 1/S \times 1/(1+r) \approx \text{price}_{\text{US goods}}$

Derivation of PPP (3)

$$e_f = \frac{1 + \text{infl}_h}{1 + \text{infl}_f} - 1$$

$\text{Infl}_h > \text{Infl}_f \Rightarrow e_f > 0$ foreign currency appreciates offsetting lower inflation

$\text{Infl}_h < \text{Infl}_f \Rightarrow e_f < 0$ foreign currency depreciates offsetting higher inflation

Example: Suppose $\text{Infl}_{\text{UK}} = 9\%$ and $\text{Infl}_{\text{US}} = 5\%$.

$$\text{Then } e_{\text{US}} = \frac{(1 + .09)}{(1 + .05)} - 1 = 3.81\%$$

Simplified PPP Relationship

When the inflation differential is small, the PPP relationship can be simplified as:

$$e_f \cong Infl_h - Infl_f$$

Example: Suppose $Infl_{UK} = 9\%$ and $Infl_{US} = 5\%$.

Then $e_{US} \cong 9 - 5 = 4\%$ similar to $1.09/1.05 - 1 = 3.8\% \approx 4\%$

UK consumers: $\Delta P_{UK} = Infl_{UK} = 9\%$
 $\Delta P_{US \text{ in } UK} = Infl_{US} + e_{US} = 9\%$

US consumers: $\Delta P_{US} = Infl_{US} = 5\%$
 $\Delta P_{UK \text{ in } US} = Infl_{UK} - e_{US} = 5\%$

Each economy can have its own inflation rate and hence its own monetary policy. The exchange rate adjusts for the difference. This is not the case between countries in the eurozone!

Problem 1

PPP and Cash Flows. Boston Co. will receive 1 million euros in one year from selling exports. It did not hedge this future transaction. Boston believes that the future value of the euro will be determined by purchasing power parity (PPP). It expects that inflation in countries using the euro will be 12% next year, while inflation in the U.S. will be 7% next year. Today the spot rate of the euro is \$1.46, and the one-year forward rate is \$1.50.

- a. Estimate the amount of U.S. dollars that Boston will receive in one year when converting its euro receivables into U.S. dollars.
- b. Today, the spot rate of the Hong Kong dollar is pegged at \$.13. Boston believes that the Hong Kong dollar will remain pegged to the dollar for the next year.

If Boston Co. decides to convert its 1 million euros into Hong Kong dollars instead of U.S. dollars at the end of one year, estimate the amount of Hong Kong dollars that Boston will receive in one year when converting its euro receivables into Hong Kong dollars.

International Fisher Effect (IFE)

- According to the Fisher effect, nominal risk-free interest rates contain **a real rate of return and anticipated inflation**.
- If all investors require the same real return, differentials in interest rates may be due to differentials in expected inflation.
- Recall that PPP theory suggests that exchange rate movements are caused by inflation rate differentials.
- The international Fisher effect (IFE) theory suggests **that currencies with higher interest rates will depreciate because higher nominal rates reflect higher expected inflation**.
- Therefore, investors hoping to capitalize on a higher foreign interest rate should earn a higher return on the investment but lose all the gain in the depreciation of the currency.

International Fisher Effect (IFE) (2)

	Country A	Country B	comment
Time preference	1 ½ %	1 ½ %	No reason for there to be a difference
+ Risk	1%	1%	We choose investments in each country that have low risk
= Real rate	2 ½ %	2 ½ %	Real rates <i>should</i> be the same
+ inflation	4 ½ %	2 ½ %	PPP $e_f = (\text{infl}_h - \text{infl}_f) = 2\%$
= interest rate (nominal rate)	7% $i_h - i_f = 2\%$	5%	The difference in inflation rates explains the difference in interest rates!

International Fisher Effect (IFE) (4)

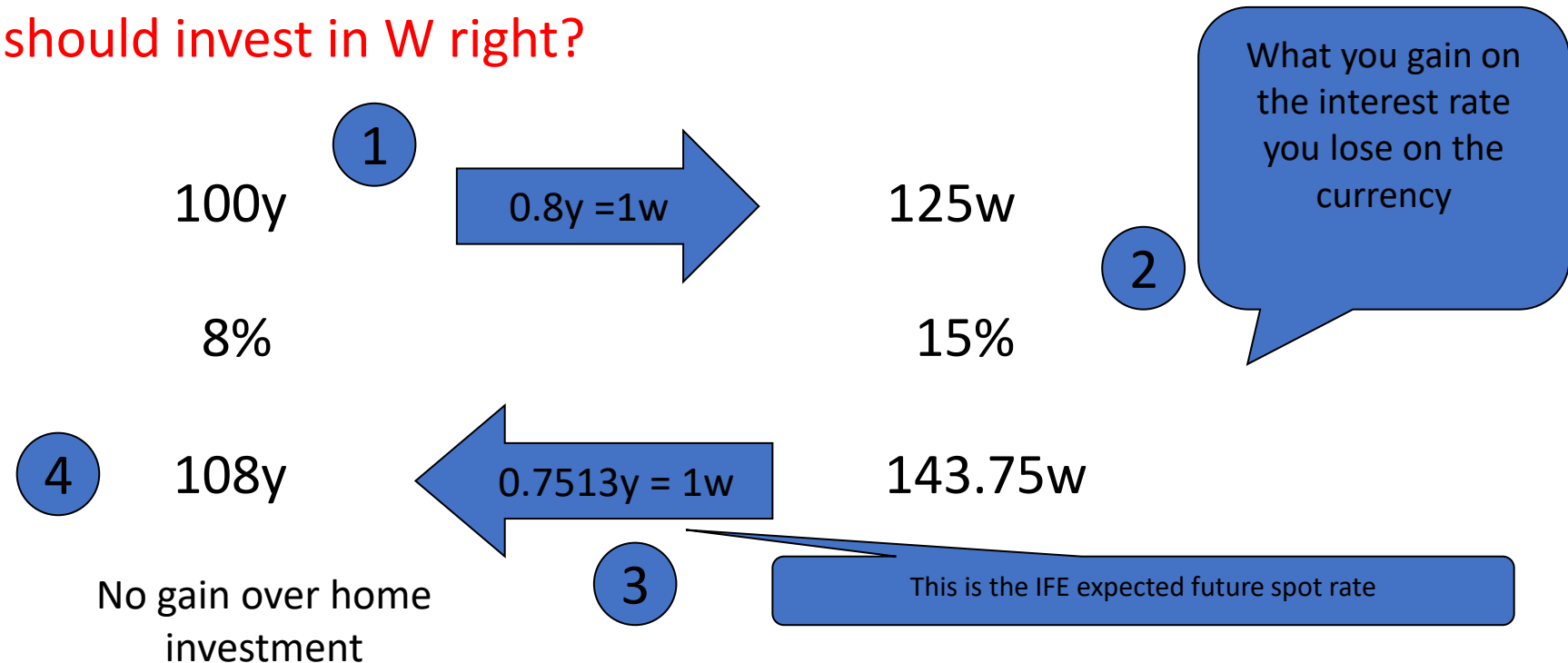
Interest rate in country W = 15%

Interest rate in country Y = 8%

Exchange rate $0.8y = 1w$

So Y investors should invest in W right?

IFE suggests



CENGAGE

For use with *International Financial Management*,
5th edn, ISBN 978-1-4737-7050-8 © Jeff Madura and Roland Fox

Why the IFE Does Not Occur

- Differences in interest rates are often small – other factors ‘**swamp**’ this influence.
- **Sovereign risk** may mean that all investments have an extra risk premium
 - $e_f = i_h - i_f$ $i_h = e_f + i_f$
 - where f is a country with economic difficulties then
 - $i_h = e_f + i_f + rp$
 - the apparent higher return is due to extra risk

Problem 2

Forecasting the Future Spot Rate Based on IFE. Assume that the spot exchange rate of the Singapore dollar is \$.70.

The one-year interest rate is 11 percent in the United States and 7 percent in Singapore.

What will the spot rate be in one year according to the IFE?

What is the force that causes the spot rate to change according to the IFE?

Interest Rate Parity

Interest rate parity

Forward rate premium p

Interest rate differential $i_h - i_f$

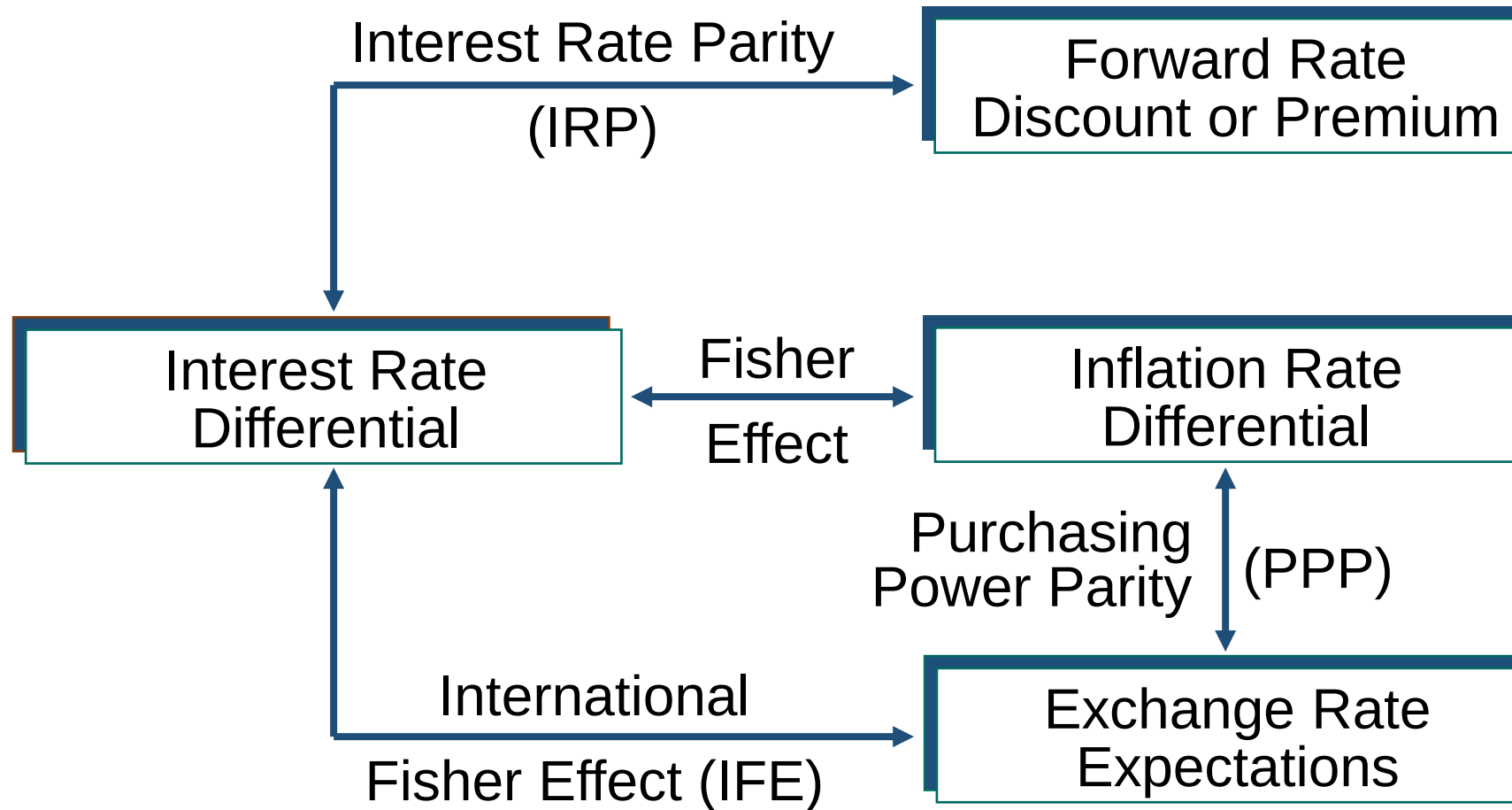
$$p = \frac{(1 + i_h)}{(1 + i_f)} - 1 \cong i_h - i_f$$

Should hold (more or less) due to the threat of arbitrage (risk free) profits.

Problem -IRP

-) **IRP.** The one-year risk-free interest rate in Mexico is 10%. The one-year risk-free rate in the U.S. is 2%. Assume that interest rate parity exists. The spot rate of the Mexican peso is \$.14.
- What is the forward rate premium?
 - What is the one-year forward rate of the peso?
 - Based on the international Fisher effect, what is the expected change in the spot rate over the next year?
 - If the spot rate changes as expected according to the IFE, what will be the spot rate in one year?
 - Compare your answers to (b) and (d) and explain the relationship.

Comparison of the IRP, PPP, and IFE Theories (1)



Comparison of the IRP, PPP, and IFE Theories (2)

Interest rate parity

Forward rate premium p

Interest rate differential $i_h - i_f$

$$p = \frac{(1 + i_h)}{(1 + i_f)} - 1 \cong i_h - i_f$$

Purchasing power parity

% Δ in spot exchange rate e_f

Inflation rate (I) differential $I_h - I_f$

$$e_f = \frac{(1 + I_h)}{(1 + I_f)} - 1 \cong I_h - I_f$$

International Fisher effect

% Δ in spot exchange rate e_f

Interest rate differential $i_h - i_f$

$$e_f = \frac{(1 + i_h)}{(1 + i_f)} - 1 \cong i_h - i_f$$

EXHIBIT 8.7 Comparison of the IRP, PPP, and IFE theories

Theory	Key Variables of Theory		Summary of Theory
Interest rate parity (IRP) or covered interest rate arbitrage	Forward rate premium (or discount)	Interest rate differential	The forward rate of one currency with respect to another will contain a premium (or discount) that is determined by the differential in interest rates between the two countries. As a result, covered interest arbitrage will provide a return that is no higher than a domestic return.
Purchasing power parity (PPP)	Percentage change in spot exchange rate	Inflation rate differential	The spot rate of one currency with respect to another will change in reaction to the differential in inflation rates between the two countries. Consequently, the purchasing power for consumers when purchasing goods in their own country will be similar to their purchasing power when importing goods from the foreign country.
International Fisher effect	Percentage change in spot exchange rate	Interest rate differential	The spot rate of one currency with respect to another will change in accordance with the differential in interest rates between the two countries. Consequently, the return on uncovered foreign money market securities will, on average, be no higher than the return on domestic money market securities from the perspective of investors in the home country.

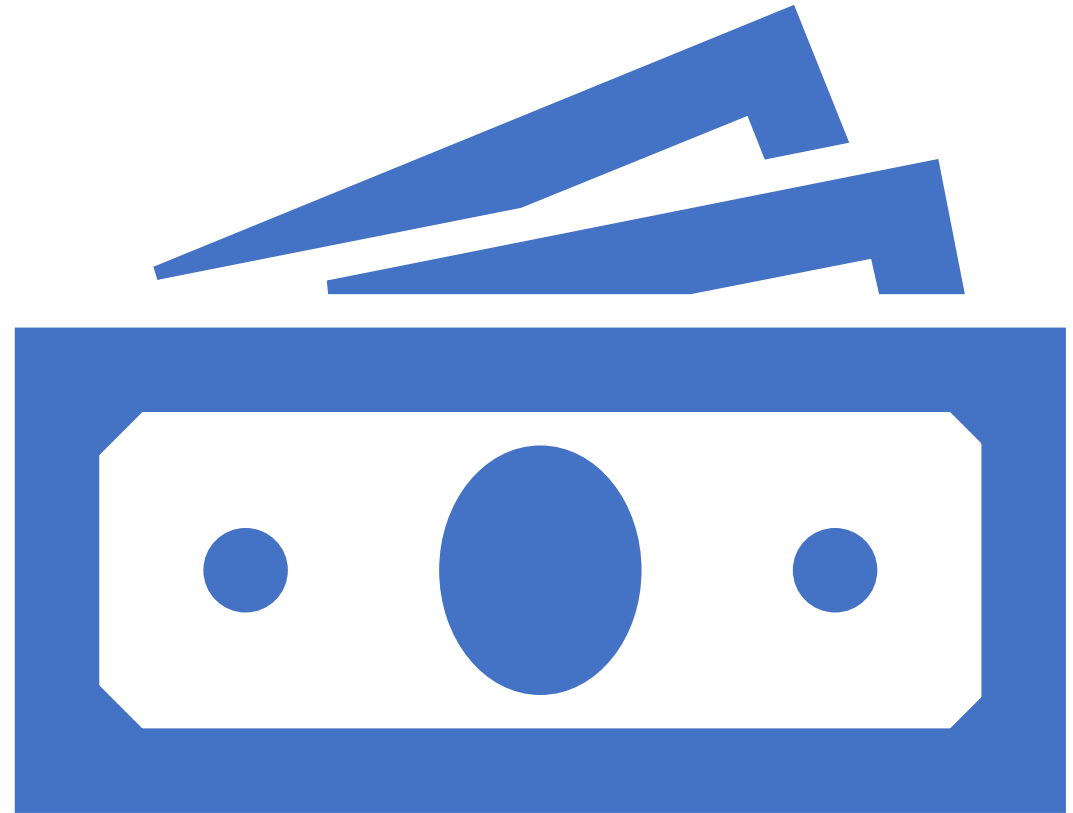
Summary 1

<https://create.kahoot.it/my-library/kahoots/d304210e-abd6-4c2b-9a98-6c519b3b6cf1>



Long term Financing

MNC



Part IV

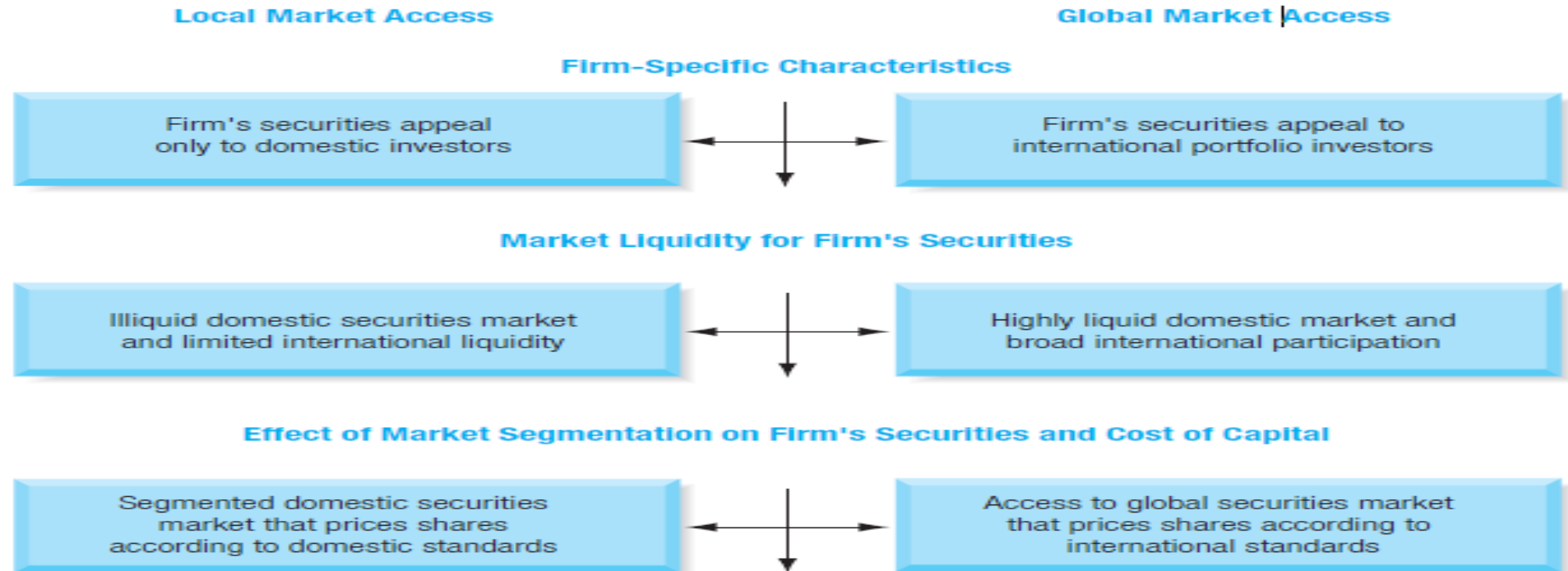
Long-Term Asset and Liability Management

Macro-level distortion

Firm specific



EXHIBIT 13.1 Dimensions of the Cost and Availability of Capital Strategy



Source: Eitmann

EXHIBIT 14.4 Equity Alternatives in the Global Market

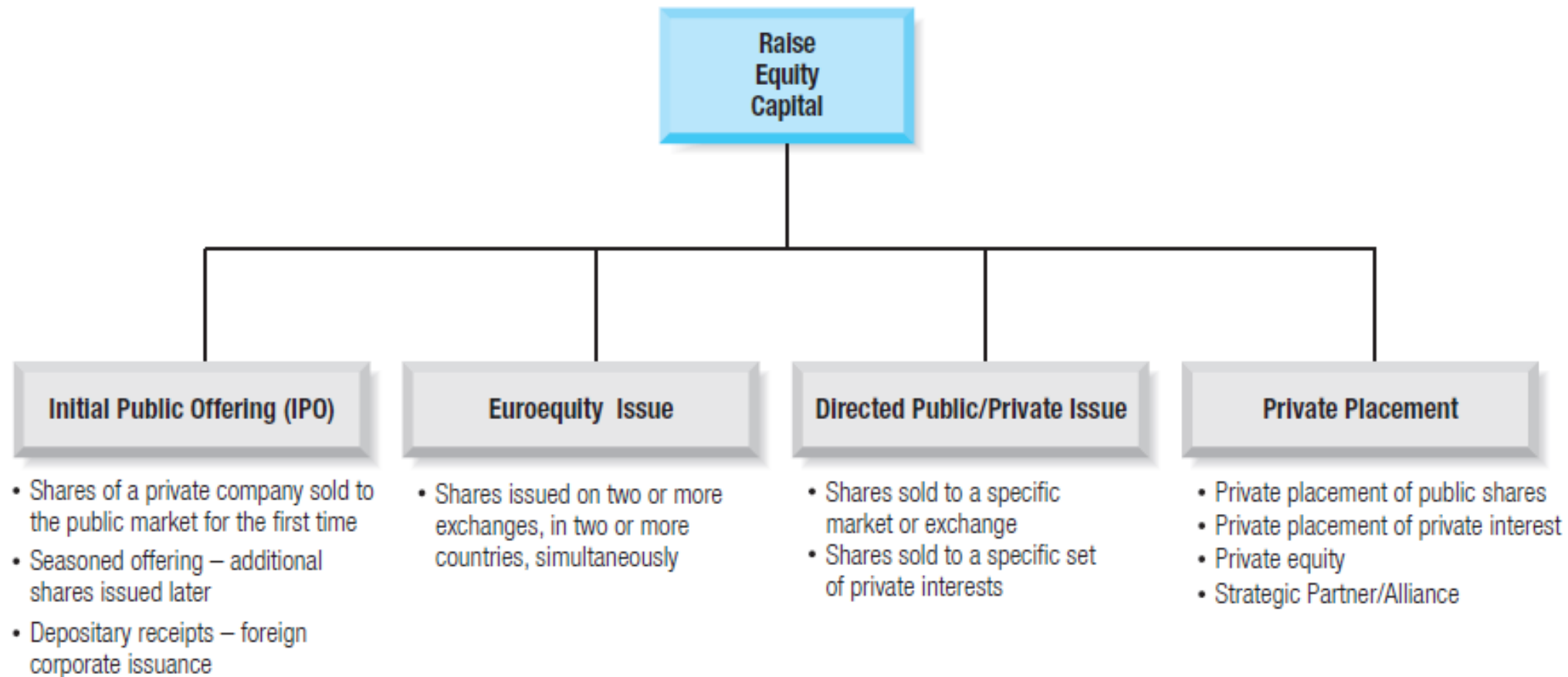


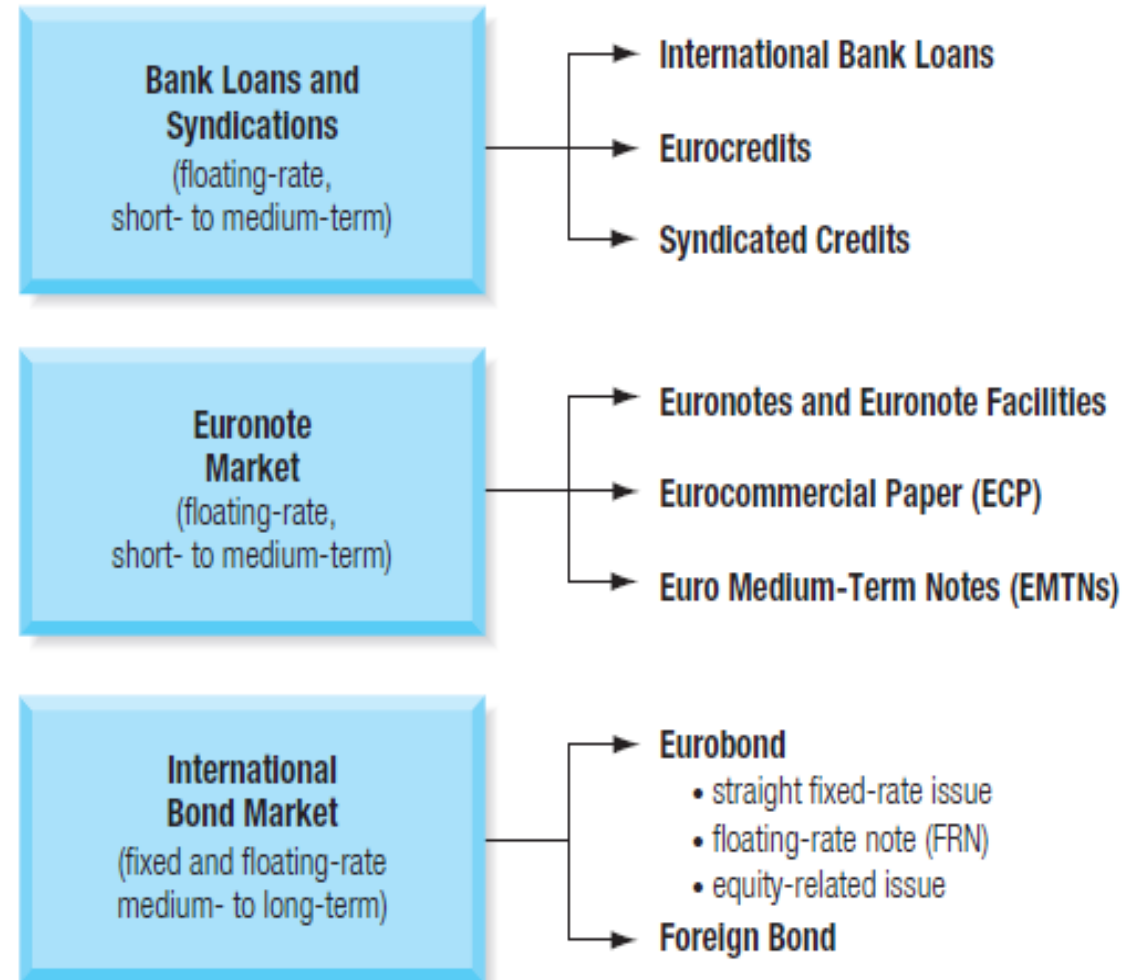
EXHIBIT 14.9**International Debt Markets and Instruments**

TABLE 14.2 Initial public offering and bond issue cost by issue size

Issue size (proceeds in \$ millions)	Total direct cost (% of proceeds)	
	IPOs	Debt
2.0–9.99	17.0	4.4
10.0–19.99	11.6	2.8
20.0–39.99	9.7	2.4
40.0–59.99	8.7	1.3
60.0–79.99	8.2	2.3
80.0–99.99	7.9	2.2
100.0–199.99	7.1	2.3
200.0–499.99	6.5	2.2
500.0 and up	5.7	1.6

- Issuance costs for debt are lower
- Debt issue costs are recurring; equity is permanent
- Debt limits flexibility more than equity

Long-Term Financing Decision

- Since MNCs commonly invest in **long-term projects**, they rely heavily on long-term financing.
- Once the capital structure decision has been made, the MNC must consider the possible sources of equity or debt, and the costs and risks associated with each source.
- **Sources of equity and debt include:**
 - domestic offering in local currency
 - global offering in multiple currencies
 - private placement to home country financial institutions
 - private placement to host country financial institutions.
- Many MNCs obtain equity funding in their home country and engage in debt financing in foreign countries.

Cost of Debt Financing

- The cost of debt financing depends on the quoted interest rate and the changes in the exchange rate of the borrowed currency over the life of the loan.
- To estimate the cost, an MNC needs to:
 - determine the amount of funds needed
 - forecast the issue price of the bond
 - forecast the exchange rates for the times when it must pay the bondholders.

Borrowing one currency and investing the money in another currency

EXAMPLE:

MNC borrowed

Volume 80 000 POUND

Interest rate 10%

Change of currency:

Devaluation 10%

$80000 * 1,1 * 0,90 = 79200$ to
repay the loan and interest

EXAMPLE:

MNC borrowed

Volume 80 000 POUND

Interest rate 4,1 %

USD Change of
currency: appreciate 2.51%

$80000 * 1,0441 * 1,0251 =$
85624 to repay the loan and
interest

EXAMPLE:

MNC borrowed

Volume 80 000 POUND

Interest rate 9.5%

INDIA Change of currency
rate :-10.5%

$80000 * 1,095 * 0,895 =$
78402 to repay the loan and
interest

Borrowing costs

UK pound cost of borrowing in different currencies in 2018

Country of currency borrowed	Change in UK pound value of foreign currency in 2018	+ Interest rate	= Actual cost over 2018
Australia	- 3.98%	5%	1.02%
France	- 1.28%	1.30%	0.02%
Germany	- 1.28%	1.70%	0.42%
India	- 10.45%	9.50%	- 0.95%
Japan	3.23%	1.50%	4.73%
Switzerland	2.03%	2.60%	4.63%
UK	0.00%	4.40%	4.40%
US	2.51%	4.10%	6.61%

Notes:

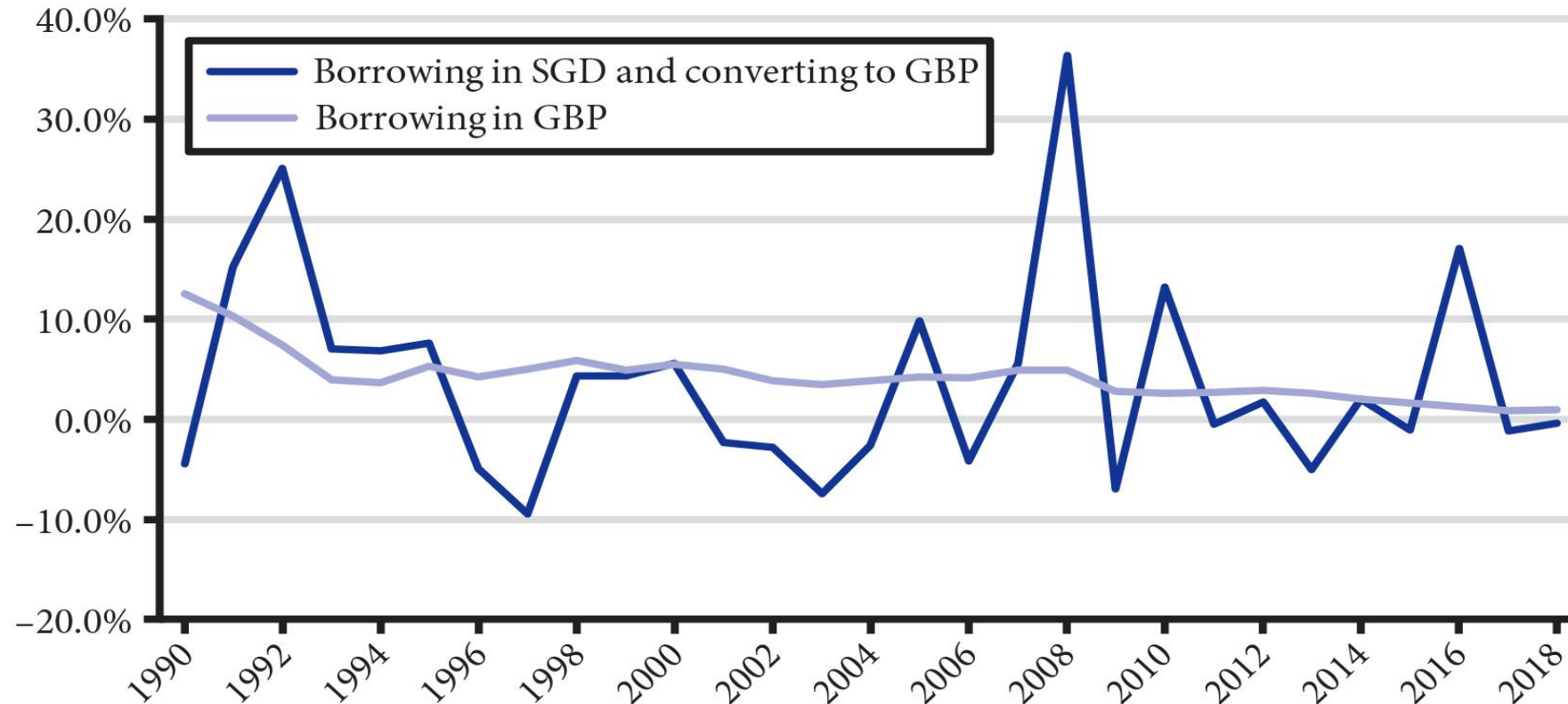
- The interest rate is an average of short-term to medium-term lending
- The scenario addressed is borrowing in the foreign currency and converting to UK pounds then repaying by converting pounds to repay the foreign loan at the end of the year. Similar effects will obtain for longer borrowing periods.

Cost of Debt Financing

- If the borrowed currency appreciates over time, an MNC will need more funds to cover the coupon or principal payments.
- The potential savings from issuing lower-yield bonds denominated in a foreign currency should be weighed against the potential risk of incurring high costs if the borrowed currency appreciates over time.

Actual Costs of Annual Financing

EXHIBIT 15.6 Borrowing in UK pounds (GBP) compared with borrowing in Singapore Dollars (SGD) and converting to GBP



Note:

- The return on foreign borrowing is influenced chiefly by changes in the exchange rate which is far more volatile than the interest rate.

EXHIBIT 16.2 Financing with bonds denominated in British pounds versus Singapore dollars

	End of Year:			Annual Cost of Financing
	1	2	3	
Financing Alternative				
(1) British pound-denominated debt (coupon rate = 14%) borrow £1 million	£140 000	£140 000	£1 140 000	14%
or				
(2) Singapore dollar-denominated bonds (coupon rate = 10%) borrow S\$3 000 000	S\$300 000	S\$300 000	S\$3 300 000	—
(2) Forecasted exchange rate of S\$	£0.3333	£0.3333	£0.3333	—
(2) Payments in pounds	£100 000	£100 000	£1 100 000	10%

Notes:

- At an exchange rate of £0.33 financing with Singapore dollars (2) is cheaper.
- The exchange rate would have to increase to $14\% / 10\% \times £0.3333 = £0.46662$ for the Singapore loan to be as expensive, this is unlikely unless inflation is higher by about 4% in the UK.

Chapter 30) to compute the forward rates.

Forward Exchange Rates. The current spot exchange rate, S , is \$1.60/£. Suppose that the yield curve in both countries is flat: The risk-free rate on dollars, $r_{\$}$, is 4%, and the risk-free interest rate on pounds, $r_{£}$, is 7%. Using the covered interest parity condition for a multi-year forward exchange rate (Eq. 30.3):

$$F_1 = S \times \frac{(1 + r_{\$})}{(1 + r_{£})} = (\$1.60/£) \frac{(1.04)}{(1.07)} = \$1.5551/£$$

$$F_2 = S \times \frac{(1 + r_{\$})^2}{(1 + r_{£})^2} = (\$1.60/£) \frac{(1.04)^2}{(1.07)^2} = \$1.5115/£$$

$$F_3 = S \times \frac{(1 + r_{\$})^3}{(1 + r_{£})^3} = (\$1.60/£) \frac{(1.04)^3}{(1.07)^3} = \$1.4692/£$$

$$F_4 = S \times \frac{(1 + r_{\$})^4}{(1 + r_{£})^4} = (\$1.60/£) \frac{(1.04)^4}{(1.07)^4} = \$1.4280/£$$

Free Cash Flow Conversion. Using these forward exchange rates, we can now calculate the expected free cash flows in dollars by multiplying the expected cash flows in pounds by the forward exchange rate, as shown in the spreadsheet in Table 31.2.

The Value of Ityesi's Foreign Project with WACC. With the cash flows of the U.K. project now expressed in dollars, we can value the foreign project as if it were a domestic U.S. project. We proceed, as we did in Chapter 18, under the assumption that the market risk of the U.K. project is similar to that of the company as a whole. As a consequence, we can use Ityesi's costs of equity and debt in the United States to calculate the WACC.²

Ityesi has built up \$20 million in cash for investment needs and has debt of \$320 million, so its net debt is $D = 320 - 20 = \$300$ million. This amount is equal to the market value of its equity, implying a (net) debt-equity ratio of 1. Ityesi intends to maintain a similar (net) debt-equity ratio for the foreseeable future. The WACC thus assigns equal weights to equity and debt (Table 31.3).

TABLE 31.2
SPREADSHEET

Expected Dollar Free Cash Flows from Ityesi's U.K. Project

	0	1	2	3	4
Dollar Free Cash Flow (\$ millions)					
1 Pound FCF (£ millions)	(17,500)	11,250	11,250	11,250	11,250
2 Forward Exchange Rate (\$/£)	1.600	1.555	1.512	1.469	1.428
3 Dollar Value of Pound FCF (1×2)	(28,000)	17,495	17,004	16,528	16,065

Reducing Exchange Rate Risk

The exchange rate risk from financing with bonds in foreign currencies can be reduced in various ways.

- Offsetting cash inflows
 - Foreign currency receipts can help offset bond payments in the same currency.
 - MNC can aggregate its cash inflows from all euro-zone countries to cover the payments for its euro-denominated bonds.
- Forward contracts
 - A firm may hedge its exchange rate risk through the forward market.
 - However, the firm may not be able to save costs due to interest rate parity.
- Currency swaps
 - A currency swap enables firms to exchange currencies at periodic intervals.
 - It can be a useful alternative to forward or futures contracts.
- Parallel loans
 - In a parallel (or back-to-back) loan, two parties simultaneously provide loans to each other (or to a subsidiary of the other party) with an agreement to repay at a specified point in the future.
- Diversifying among currencies
 - A firm may issue bonds in several foreign currencies for diversity.
 - To avoid the higher transaction costs associated with multiple bond issues, the firm may develop a currency cocktail bond.
 - One popular currency cocktail is the Special Drawing Right (SDR).

Interest Rate Risk from Debt Financing

- An MNC must also decide on the maturity that it should use for its debt.
- If the bond term is too short, the MNC may have to refinance at a higher interest rate.
- However, if the bond term matches the expected business life, the MNC is obligated to continue paying interest at the same rate even when market interest rates fall.
- Before making the debt maturity decision, MNCs may want to assess the yield curves of the countries in which they need funds.
- A yield curve is shaped by the demand for and supply of funds at various maturity levels in a country's debt market.
- An upward-sloping yield curve means that the annualized yields are lower for short-term debt than for long-term debt.
- MNCs that wish to issue a long-term bond but want to avoid the prevailing fixed rate may consider floating rate bonds.
- For example, the coupon rate is frequently tied to the London Interbank Offer Rate (LIBOR).
- If the coupon rate is floating, forecasts are required for both exchange rates and interest rates.

Hedging with Interest Rate Swaps (1)

- When MNCs issue floating-rate bonds that expose them to interest rate risk, they may use interest rate swaps to hedge the risk.
- Interest rate swaps enable a firm to exchange fixed rate payments for variable rate payments, and vice versa.
- Bond issuers use swaps to reconfigure their future cash flows in a way that offsets their payments to bondholders.
- Financial intermediaries are usually involved in swap agreements. They match up participants and assume the default risk involved for a fee.
- In a plain vanilla swap, the floating rate payer is typically highly sensitive to interest rate changes and seeks to reduce interest rate risk.

Cost of Capital

CAPM model

Capital Asset Pricing Model

Here is the formula:

$$R_a = R_{rf} + \beta_a * (R_m - R_{rf})$$

where:

R_a = Expected return on a security

R_{rf} = Risk-free rate

R_m = Expected return of the market

β_a = The beta of the security

$(R_m - R_{rf})$ = Equity market premium

The Cost of Capital

The Cost of Capital

A domestic firm normally finds its *cost of capital* by evaluating where and from whom it will raise its capital. The cost will obviously differ on the mix of investors interested in the firm, investors willing and able to buy its equity shares, and the debt available to the firm, raised from the domestic bank and debt market.

The firm then calculates its *weighted average cost of capital* (WACC) by combining the cost of equity with the cost of debt in proportion to the relative weight of each in the firm's optimal long-term financial structure. More specifically,

$$k_{WACC} = k_e \frac{E}{V} + k_d (1 - t) \frac{D}{V}$$

weighted average after-tax cost of capital

k_e = risk-adjusted cost of equity

k_d = before-tax cost of debt

t = marginal tax rate

E = market value of the firm's equity

D = market value of the firm's debt

V = market value of the firm's securities ($D + E$)

Cost of capital calculation

Maria assumes the risk-free rate of interest (k_{rf}) as 4%, using the U.S. government 10-year Treasury bond rate. The expected rate of return of the market portfolio (k_m) is assumed to be 9%, the expected rate of return on the market portfolio held by a well-diversified domestic investor. Trident's estimate of its own systematic risk, its beta, against the domestic portfolio is 1.2. Trident's cost of equity is then

$$k_e = k_{rf} + \beta (k_m - k_{rf}) = 4.00\% + 1.2 (9.00\% - 5.00\%) = 10.00\%.$$

Trident's cost of debt (k_d) the before tax cost of debt estimated by observing the current yield on Trident's outstanding bonds combined with bank debt, is 8%. Using 35% as the corporate income tax rate for the United States, Trident's after-tax cost of debt is then

$$k_d (1 - t) = 8.00 (1 - 0.35) = 8.00 (0.65) = 5.20\%.$$

Trident's long-term capital structure is 60% equity (E/V) and 40% debt (D/V), where V is Trident's total market value. Trident's weighted average cost of capital k_{WACC} is then

$$k_{WACC} = k_e \frac{K}{V} + k_d (1 - t) \frac{D}{V} = 10.00\% (.60) + 5.20\% (.40) = 8.08\%$$

This is Trident's cost of capital using the traditional domestic CAPM estimate of the cost of equity.

A second calculation of Trident's cost of equity, this time using the ICAPM, yields different results. Trident's beta, when calculated against a larger global equity market index which includes these foreign markets and their investors, is lower, 0.90. The expected market return for a larger globally integrated equity market is a lower value as well, 8.00%. The ICAPM cost of equity is a much lower value of 7.60%.

$$k_e^{\text{global}} = k_{rf}^g + \beta_j^g (k_m^g - k_{rf}^g) = 4.00\% + 0.90 (8.00\% - 4.00\%) = 7.60\%$$

Maria now recalculates Trident's WACC using the ICAPM estimate of equity costs, assuming the same debt and equity proportions and the same cost of current debt. Trident's WACC is now estimated at a lower cost of 6.64%.

$$k_{WACC}^{\text{ICAPM}} = k_e^{\text{global}} \frac{E}{V} + k_d (1 - t) \frac{D}{V} = 7.60\% (.60) + 5.20\% (.40) = 6.64\%$$

Maria believes that this is a more appropriate estimate of Trident's cost of capital. It is fully competitive with Trident's main rivals in the telecommunications hardware industry segment worldwide, which are mainly headquartered in the United States, the United Kingdom, Canada, Finland, Sweden, Germany, Japan, and the Netherlands. The key to Trident's favorable global cost and availability of capital going forward is its ability to attract and hold the international portfolio investors that own its stock.

TABLE 31.3Ityesi's Current Market Value Balance Sheet (\$ million)
and Cost of Capital without the U.K. Project

Assets		Liabilities		Cost of Capital	
Cash	20	Debt	320	Debt	6%
Existing Assets	600	Equity	300	Equity	10%
	620		620		

With Ityesi's cost of equity at 10% and its cost of debt at 6%, we calculate Ityesi's WACC as follows:

$$\begin{aligned}
 r_{wacc} &= \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - \tau_D) \\
 &= (0.5)(10.0\%) + (0.5)(6.0\%)(1 - 40\%) = 6.8\%
 \end{aligned}$$

We can now determine the value of the foreign project, including the tax shield from debt, by calculating the present value of the future free cash flows using the WACC:

$$\frac{17.495}{1.068} + \frac{17.004}{1.068^2} + \frac{16.528}{1.068^3} + \frac{16.065}{1.068^4} = \$57.20 \text{ million}$$

Because the upfront cost of launching the product line in dollars is only \$28 million, the net present value is $57.20 - 28 = \$29.20$ million. Thus, Ityesi should undertake the U.K. project.

Using the Law of One Price to Determine Debt Tax Credit

Problem

Camacho Enterprises is a U.S. company that is considering expanding by acquiring Xtapa, Inc., a firm in Mexico. The acquisition is expected to increase Camacho's free cash flows by 21 million pesos the first year; this amount is then expected to grow at a rate of 8% per year. The price of the investment is 525 million pesos, which is \$52.5 million at the current exchange rate of 10 pesos/\$. Based on an analysis in the Mexican market, Camacho has determined that the appropriate after-tax peso WACC is 12%. If Camacho has also determined that its after-tax dollar WACC for this expansion is 7.5%, what is the value of the Mexican acquisition? Assume that the Mexican and U.S. markets for risk-free securities are integrated and that the yield curve in both countries is flat. U.S. risk-free interest rates are 6%, and Mexican risk-free interest rates are 9%.

Solution

Let's begin by calculating the NPV of the expansion in pesos and converting the result into dollars at the spot rate. Putting the free cash flows on a timeline:



The net present value of these cash flows at the peso WACC is

$$NPV = \frac{21}{0.12 - 0.08} - 525 = 0$$

so the purchase is a zero-NPV transaction. Presumably, Camacho is competing with other Mexican companies for the purchase, and they have bid up the price to the point that $NPV = 0$.

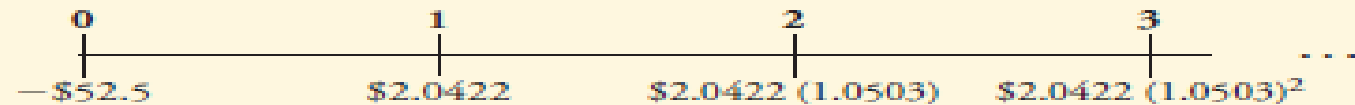
We can also compute the NPV in dollars by converting the expected cash flows into dollars using forward rates. The N -year forward rate (Eq. 30.3 in Chapter 30) expressed in pesos/\$ is

$$F_N = S \times \frac{(1 + r_p)^N}{(1 + r_s)^N} = 10 \times \left(\frac{1.09}{1.06} \right)^N = 10 \times 1.0283^N = 10.283 \times 1.0283^{N-1}$$

Thus, the dollar expected cash flows are the peso cash flows (from the earlier timeline) converted at the appropriate forward rate (we divide by the forward rate because it is in pesos/\$):

$$C_p^N / F_N = \frac{21(1.08)^{N-1}}{10.283 \times 1.0283^{N-1}} = 2.0422 \times 1.0503^{N-1}$$

The dollar expected cash flows are therefore



so the dollar cash flows grow at about 5% per year. The NPV of these cash flows is

$$NPV = \frac{2.0422}{0.075 - 0.0503} - 52.5 = \$30.18 \text{ million}$$

Which NPV more accurately represents the benefits of the expansion? The answer depends on the source of the difference. To compute the dollar expected cash flows by converting the peso expected cash flows at the forward rate, we must accept the assumption that spot rates and the project cash flows are uncorrelated. The difference might simply reflect that this assumption failed to hold. Another possibility is that the difference reflects estimation error in the respective WACC estimates.

If Camacho is relatively confident in its assumptions about spot rates and its WACC estimates, a third possibility is that Mexican and U.S. capital markets are not integrated. In this case, Camacho, because of its access to U.S. capital markets, might have a competitive advantage. Perhaps other companies with which it is competing for the purchase of Xtapa are all Mexican firms that do not have access to capital markets outside of Mexico. Hence Camacho can raise capital at a cheaper rate. Of course, this argument also requires that other U.S. companies are not competing for the purchase of Xtapa. Camacho, however, might have special knowledge of Xtapa's markets that other U.S.-based companies lack. This knowledge would give Camacho a competitive advantage in the product market over other U.S. companies and puts it on an equal footing in the product market with other Mexican companies. Because it has a competitive advantage in capital markets over other Mexican companies, the NPV of the purchase is positive for Camacho, but zero for the other bidders for Xtapa.

Thank you
for your
attention!

